

Free-Tier LLM Alternatives to Running Locally

Running a model locally with `llama.cpp` / GPT4All requires decent hardware. If your machine doesn't have enough RAM or CPU power, the options below are **free to start**, require no credit card for most, and produce better results than the small local models.

Quick Comparison

Provider	Recommended Model	Free Tier	Best For
Groq	<code>llama-3.1-8b-instant</code>	Yes — generous rate limits	Speed, DX, OpenAI-compatible
Google Gemini (AI Studio)	<code>gemini-2.0-flash</code>	Yes — no credit card needed	Quality, system prompts, multimodal
OpenRouter	Many (Llama, Mistral, Gemma...)	Yes — free models available	Benchmarking multiple models easily
Mistral AI	<code>mistral-small-latest</code>	Yes — free tier via la Plateforme	European data residency, GDPR angle
Cloudflare Workers AI	Llama 3, Phi	Yes — 10k neurons/day free	Edge deployment, novel architecture
Hugging Face Inference API	Hundreds of open models	Yes — rate-limited	Exploring open-source models
Cohere	<code>command-r</code>	Yes — trial tier	RAG, document Q&A use cases

Recommended Picks

1. Groq — Best for speed and developer experience

- Sign up at <https://console.groq.com>
- Get an API key instantly, no credit card
- Uses the **OpenAI-compatible** chat completions format
- `llama-3.1-8b-instant` is fast and capable enough for most chatbot use cases

```
import Groq from 'groq-sdk'

const groq = new Groq({ apiKey: process.env.GROQ_API_KEY })

const response = await groq.chat.completions.create({
  model: 'llama-3.1-8b-instant',
  messages: [
    { role: 'system', content: 'You are a helpful assistant.' },
    { role: 'user', content: userMessage }
  ]
})

console.log(response.choices[0].message.content)
```

Install: `npm install groq-sdk`

2. Google Gemini (AI Studio) — Best quality on free tier

- Sign up at <https://aistudio.google.com>
- gemini-2.0-flash is free with generous quotas
- Supports system prompts, streaming, and multimodal input
- Uses Google's own SDK (@google/generative-ai)

```
import { GoogleGenerativeAI } from '@google/generative-ai'

const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY)
const model = genAI.getGenerativeModel({
  model: 'gemini-2.0-flash',
  systemInstruction: 'You are a helpful assistant.'
})

const result = await model.generateContent(userMessage)
console.log(result.response.text())
```

Install: `npm install @google/generative-ai`

3. OpenRouter — Best for benchmarking multiple models

- Sign up at <https://openrouter.ai>

- Free models include: `meta-llama/llama-3.1-8b-instruct:free`, `mistralai/mistral-7b-instruct:free`, `google/gemma-3-4b-it:free`
- **Fully OpenAI-compatible** — just change the base URL
- Ideal for the **"GenAI performances" grading criterion**: swap models by changing one line

```
import OpenAI from 'openai'

const client = new OpenAI({
  baseURL: 'https://openrouter.ai/api/v1',
  apiKey: process.env.OPENROUTER_API_KEY,
})

const response = await client.chat.completions.create({
  model: 'meta-llama/llama-3.1-8b-instruct:free', // swap this line to
  benchmark
  messages: [
    { role: 'system', content: 'You are a helpful assistant.' },
    { role: 'user', content: userMessage }
  ]
})
```

Install: `npm install openai`

4. Mistral AI — Interesting for the Data Privacy section (GDPR)

- Sign up at <https://console.mistral.ai>
- Free tier available (`mistral-small-latest`)
- French company, EU data residency — directly relevant to the **Data Privacy (4pts)** section of the project
- OpenAI-compatible API

```
import MistralClient from '@mistralai/mistralai'

const client = new MistralClient(process.env.MISTRAL_API_KEY)

const response = await client.chat({
  model: 'mistral-small-latest',
  messages: [{ role: 'user', content: userMessage }]
})
```

Install: `npm install @mistralai/mistralai`

5. Cloudflare Workers AI — Interesting for the Innovation section

- Free tier: 10,000 neurons/day
 - Deploy your chatbot **on the edge**, no server needed
 - Interesting architecture angle for the **Conception & Innovation** sections
 - See: <https://developers.cloudflare.com/workers-ai/>
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Switching Between Providers

Most providers use the **OpenAI-compatible format** (`/v1/chat/completions`). You can abstract the provider behind an environment variable:

```
// .env
LLM_PROVIDER=groq           # or: openrouter, mistral
LLM_MODEL=llama-3.1-8b-instant
```

This makes it easy to benchmark multiple models — which is exactly what the **"Working on gen AI performances"** (4pts) criterion rewards.

Tips for the Grading Criteria

Criterion	How these alternatives help
GenAI integration (2pts)	Any of the above satisfies this — pick whichever is easiest to set up
GenAI performances (4pts)	Use OpenRouter to test 2–3 models with the same prompt, document the differences
Data privacy (4pts)	Mistral (EU-based) vs OpenAI (US-based) is a concrete real-world comparison to write about
Innovation (4pts)	Cloudflare Workers AI edge deployment is a less common architecture worth highlighting